# IQRA IRFAN

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## EDUCATION

***University of California, Los Angeles (UCLA)*** ***June 2022***

B.S. Human Biology and Society, ***Highest Departmental Honors***

***California State University, Los Angeles (CSULA)*** ***December 2024***

Computer Science Post-Baccalaureate

***University of Southern California (USC)*** ***January 2025 - Present***

M.S. Computer Science

## RELEVANT PROFESSIONAL EXPERIENCES

**Software Analyst at Cliffwater Corporate Lending Fund - Full-Time** ***September 2024 - November 2025***

- Promoted to Software Analyst after completion of fall internship. Roles include software development, data analytics, machine learning models, artificial intelligence observability, information security controls, open source API integration, robotic process automation prototyping, and other duties in support of the firm's investment advisory services.

**Full-Stack Mobile Application Software Engineer at 7000 Languages - Junior** ***June 2023 - Present***

- Promoted to junior-level full-stack mobile application software engineer after completion of spring internship. Introduced a dynamic activity bar and an interactive home page using React Native, TypeScript, and Python, optimizing performance and enhancing user experience.
- Led UI enhancements through strategic refactoring, significantly improving the application's user interface. Received positive feedback from users, reporting increased learning outcomes.

**Mobile Application Software Development Spring Internship at 7000 Languages - Intern** ***January 2023 - June 2023***

- Built a cross-platform iOS/Android app using React Native, TypeScript, and Python to support creation of indigenous language learning courses. Improved app stability and implemented offline data storage with MongoDB/Realm in an agile team environment.

## RECENT SOFTWARE PROGRAMMING AND RESEARCH PROJECTS

- **Artificial Intelligence & Robotics Research – Modular & Adaptive Systems Lab, USC**
  - Researching self-reconfigurable modular robots in the Modular & Adaptive Systems Lab, inspired by USC's SuperBot project and focused on AI-driven adaptive control and locomotion. Implementing embedded systems integration and microcontroller programming for autonomous, environment-responsive robotic behavior.
- **Reinforcement Learning Join Order Optimizer - PostgreSQL** ***October 2025***
  - Engineered a Reinforcement Learning–based join order optimizer in PostgreSQL, inspired by Marcus et al., "Bao: Learning to Steer Query Optimizers" (VLDB'19), to improve real-time query planning efficiency.
  - Integrated a Deep Q-Learning agent via PostgreSQL planner hooks and benchmarked on TPC-H workloads, achieving measurable reductions in query optimization latency compared to the native cost-based optimizer.
- **Automated Private Investments Daily Financial Statement Production (Cliffwater)** ***August 2025***
  - Developed a C# automation system to generate and distribute daily investor statements for three funds. Integrated UMB data retrieval, encrypted SQL storage, and PDF generation via HTML, automated through Docker, Kubernetes, Azure logging, and OpsGenie alerts.

## SKILLS AND QUALIFICATIONS

- **Frontend:** JavaScript, TypeScript, ReactJS, React Native, C, JavaFX, HTML/CSS
- **Backend:** Java, Python, NextJS, NodeJS, C#
- **Database management:** MongoDB, AWS, MySQL, Firebase, PostgreSQL, Query Optimization and Planning
- **Testing & deployment:** XCode, TestFlight, Python Unittest, React Testing Library, Apple App Store, Google Play Store
- **Version Control:** Bitbucket, Git and Github
- **Machine Learning & AI:** Reinforcement Learning, Deep Q-Learning, Query Optimization, Computer Vision, CNNs, TensorFlow, Model Training & Fine-Tuning
- **Finance & Data:** Bloomberg Terminal, Custom Stock Ticker Creation, Quantitative Financial Data Analysis